

Automated Tuning of End-to-end Neural Flight Controllers for Autonomous Nano-drones

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Context

- Autonomous UAVs are becoming more popular due to their capabilities of performing surveillance, inspection or interact with smart agents.
- UAVs can be classified by their weight.
- This work focuses on autonomous nano-UAVs.



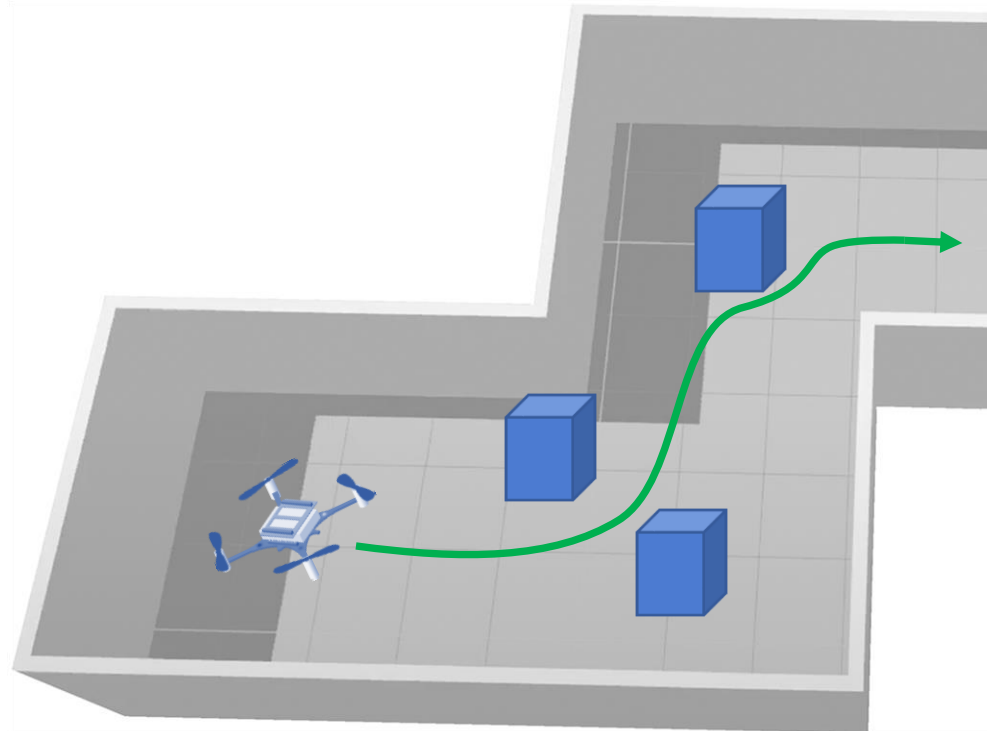
[1] A. Loquercio et al., "DroNet: Learning to Fly by Driving," 2018

Vehicle Class	⌀ : Weight [cm:kg]	Power [W]	Onboard Device
<i>std-size</i> [11]	$\sim 50 : \geq 1$	≥ 100	Desktop
<i>micro-size</i> [12]	$\sim 25 : \sim 0.5$	~ 50	Embedded
<i>nano-size</i> [13]	$\sim 10 : \sim 0.01$	~ 5	MCU
<i>pico-size</i> [14]	$\sim 2 : \leq 0.001$	~ 0.1	ULP

[2] D. Palossi et al., "A 64mW DNN-based Visual Navigation Engine for Autonomous Nano-Drones," 2019

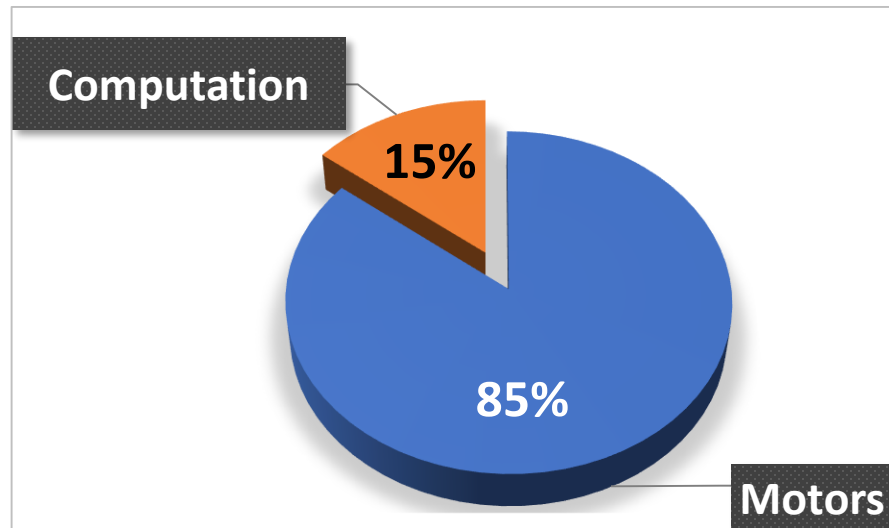
Context

- Nano-UAVs are the ideal candidates for autonomous indoor navigation.
- They can safely operate near humans and reach narrow spots with their reduced dimensions.



Problem

- Total power envelope of a few Watts, of which only 5-15% is available for the onboard computation [1].
- Computational unit is on-board the nano-UAV.
- Incapable of real-time execution of complex navigation algorithms.

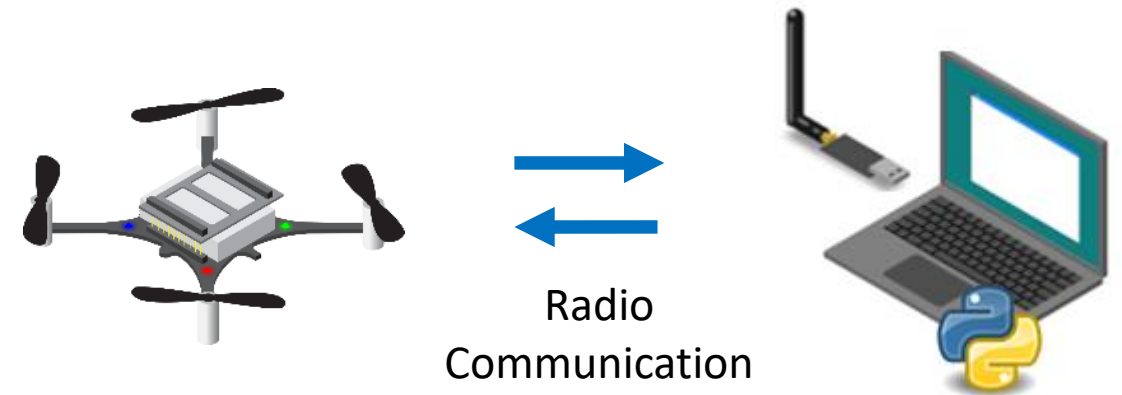
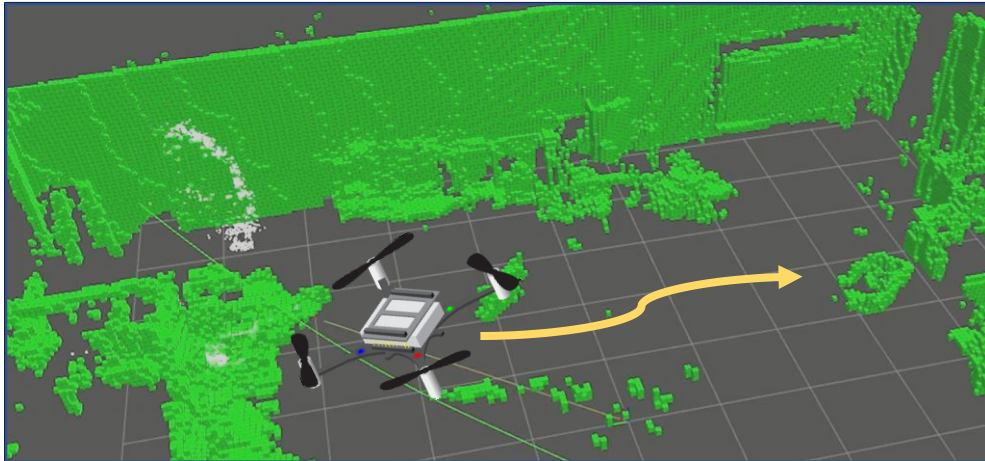


**Reduced computational
resources**



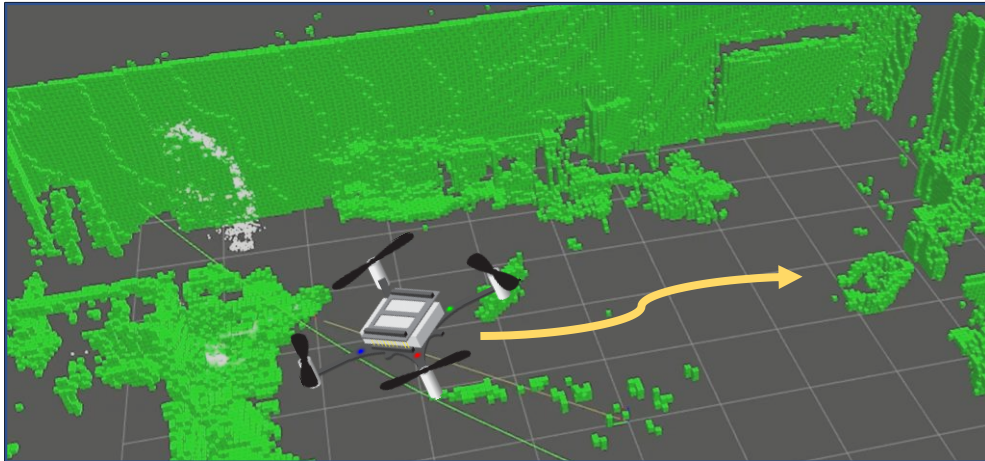
Autonomous Navigation: Classical Approaches

- SLAM creates a map of the environment and plans the trajectory according to it
→ unfeasible on a low-resources MCU.
- SLAM usually runs in powerful embedded computers (i.e., Nvidia Jetson TX2 [1]).
- Possible solution: off-load the computation to a base-station.

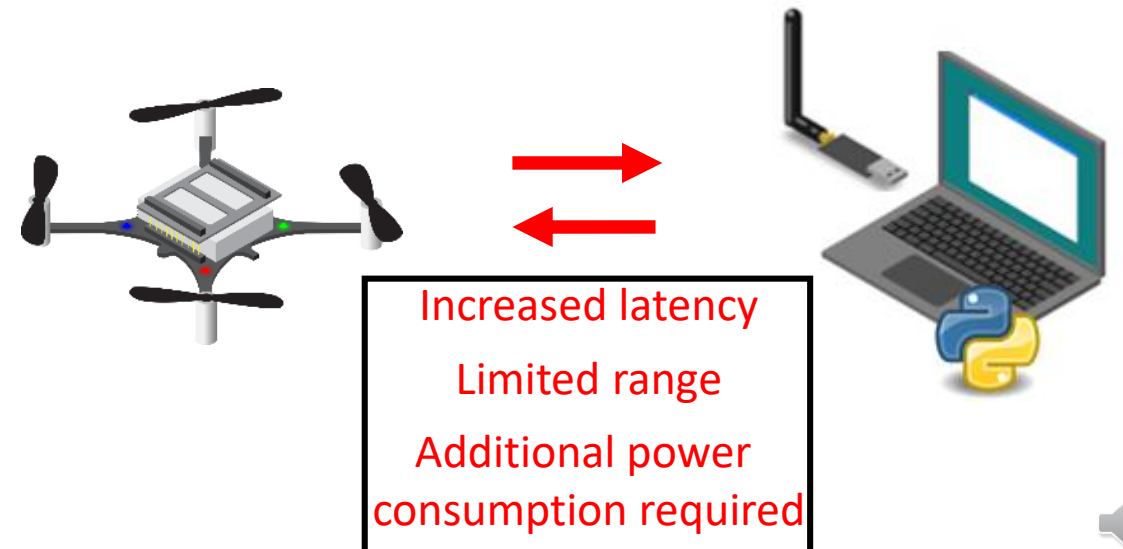


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- SLAM usually runs in powerful embedded computers (i.e., Nvidia Jetson TX2 [1]).
- Possible solution: off-load the computation to a base-station.
- This comes with additional drawbacks

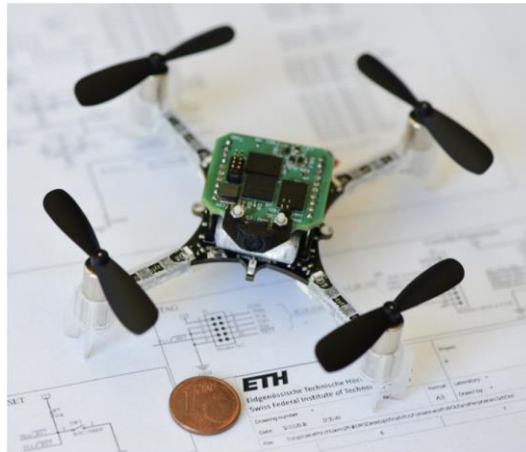


[1] T. Peng et al., *An Evaluation of Embedded GPU Systems for Visual SLAM Algorithms*, 2020.



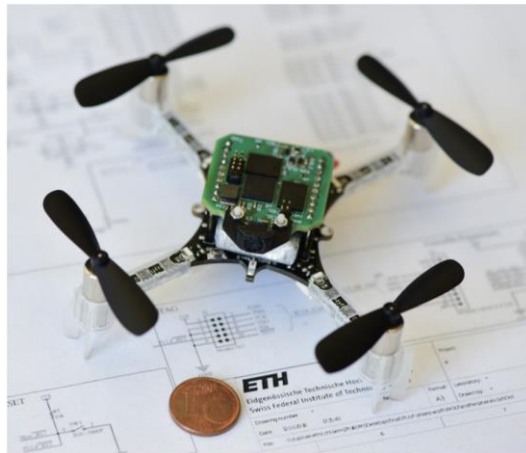
Solution

- End-to-end CNNs proved their capabilities to enable autonomous navigation.
- SoA: PULP-DroNet [1] enables nano-UAV navigation - 40 MMAC/frame and 10 frame/s.
- GAP8: modern MCU based on the PULP architecture - 1GMAC/s.



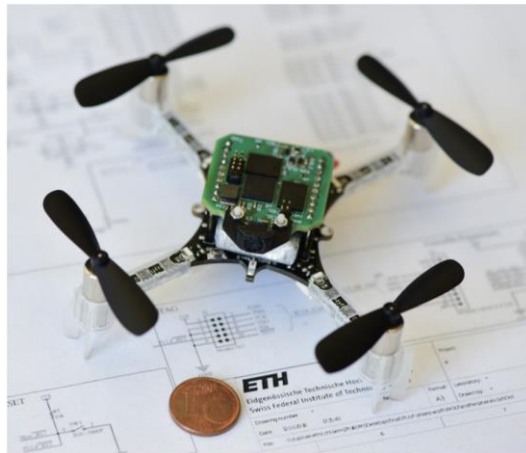
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- **Challenge: Complexity reduction methods are mostly hand-crafted.**



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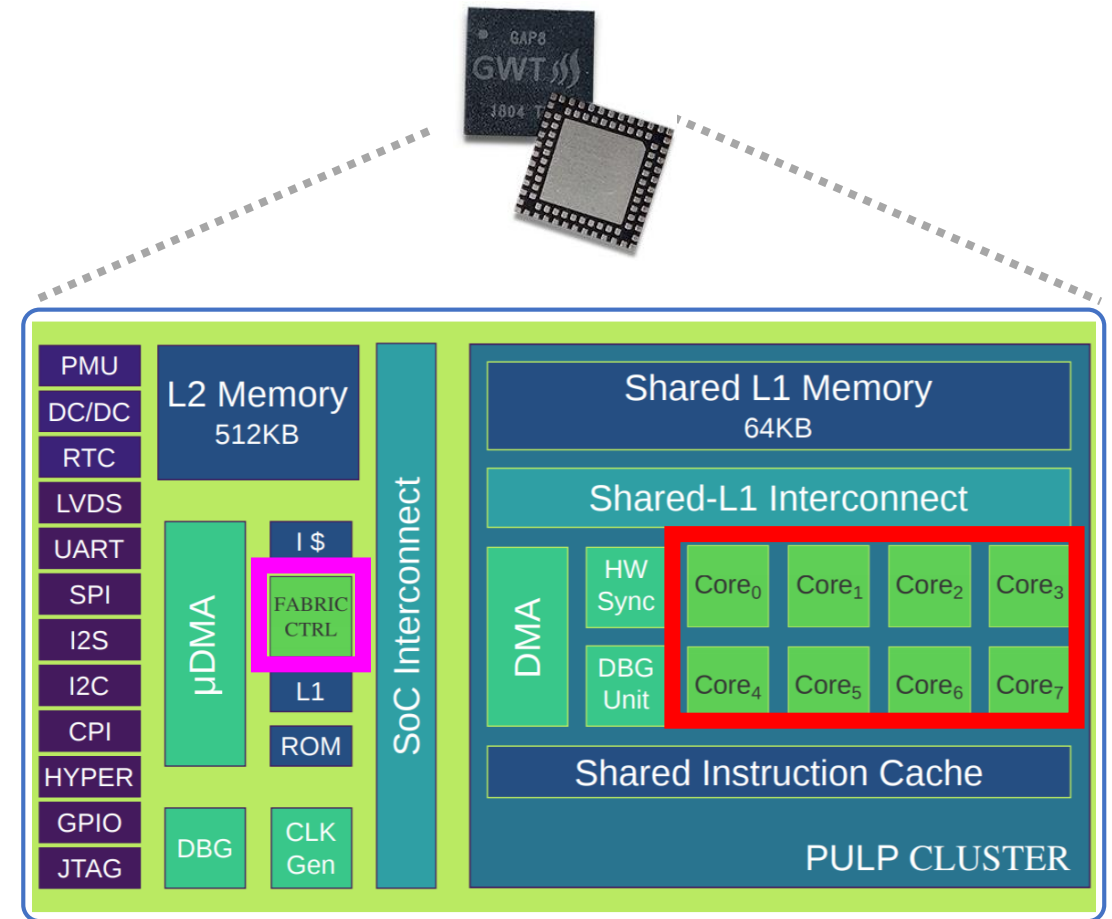
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- **Challenge: Complexity reduction methods are mostly hand-crafted.**
- **Contribution: evaluation of automatic hardware-aware deployment tools.**



Platform

GAP8 SoC (GreenWaves Technologies)

- PULP architecture (parallel-ultra-low-power)
- 1 main core (**Fabric Controller**)
- 8-cores parallel **Cluster**
- Memory: L1 (64KB), L2 (512KB)
L3 (8MB SRAM or 64MB Flash)
- Throughput > 1 GMAC/s



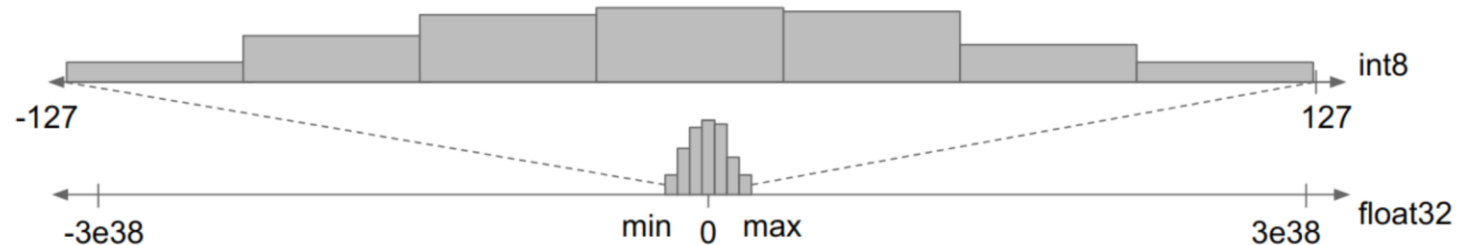
Source [1]



Running a CNN on GAP8

Quantization:

- Remaps the parameters' numerical representation from float32 to int8.



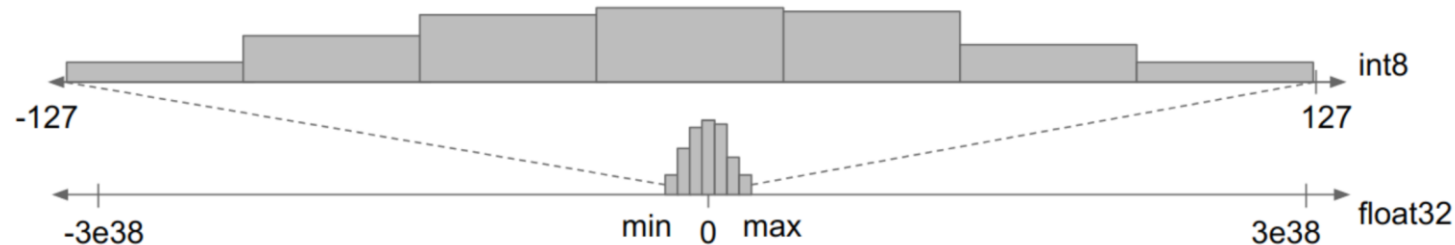
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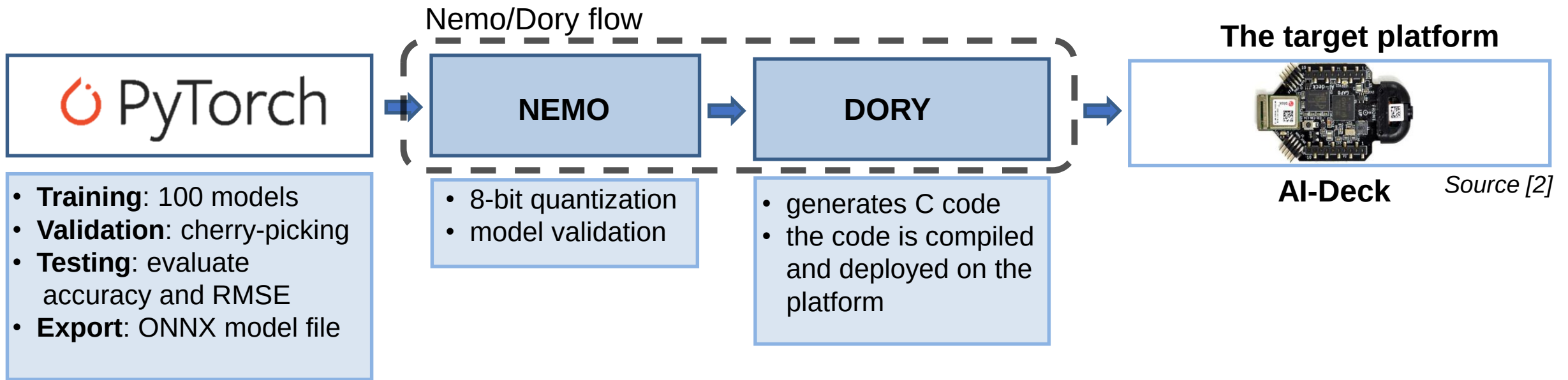
Source [1]

Deployment:

- Generates the C code out of the quantized CNN model that:
- Maximizes the parallel execution over all available cores.
- Minimizes the data transfers overhead.



The Nemo/Dory Tool-set

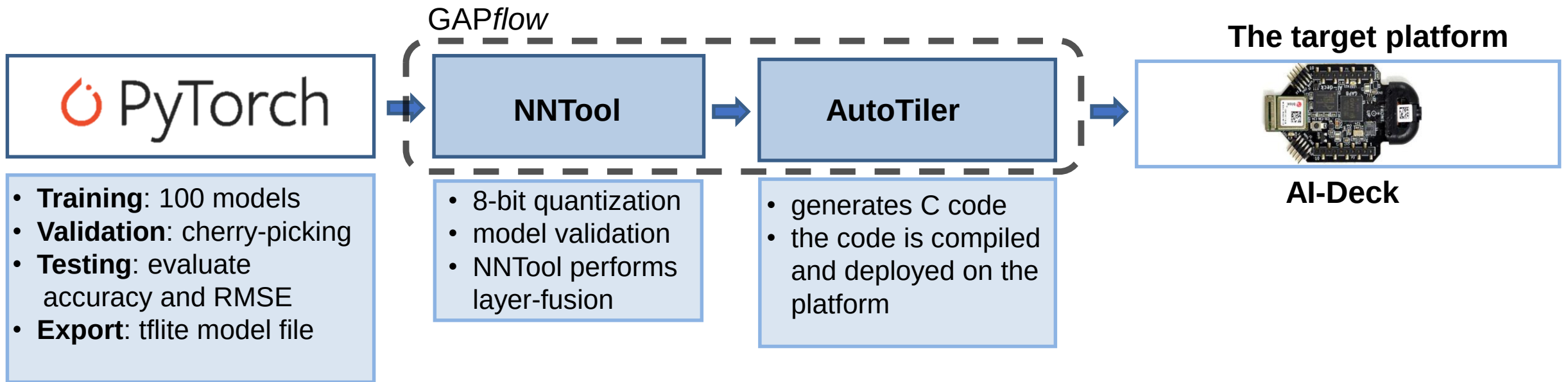


[1] A. Burrello et al., "Dory: Automatic end-to-end deployment of realworld dnns on low-cost iot mcus," 2020

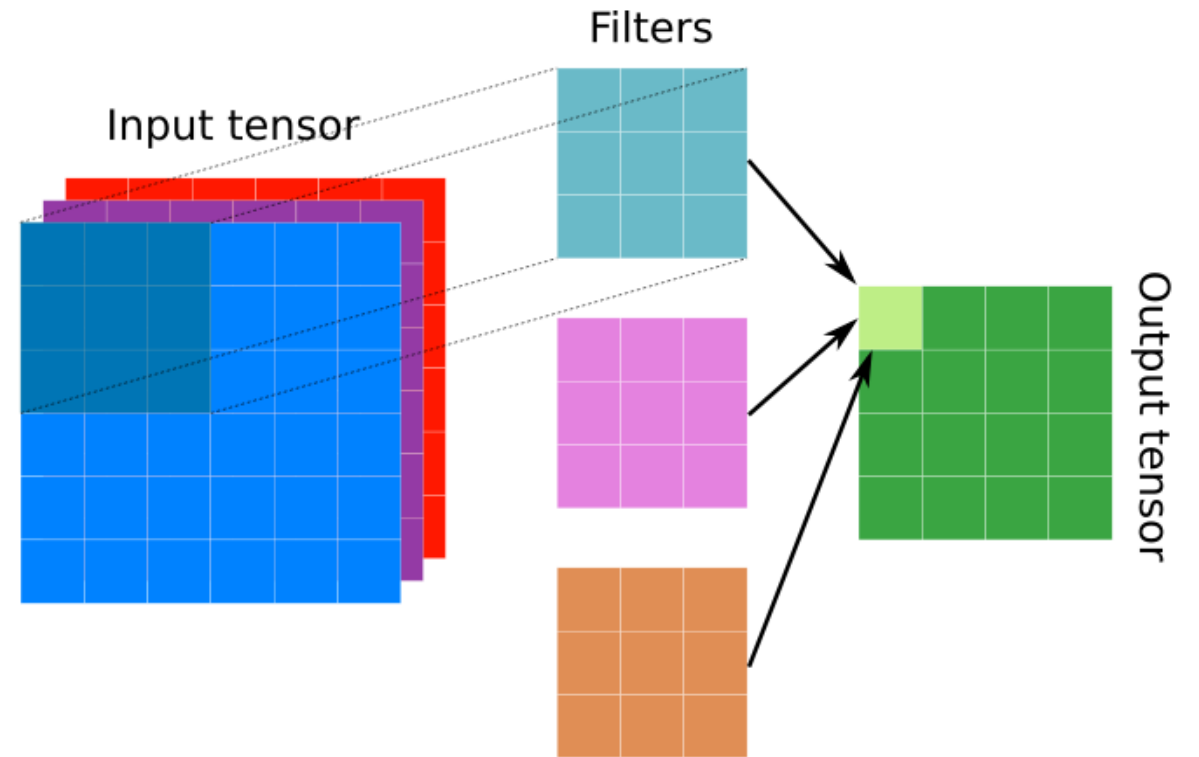
[2] <https://www.bitcraze.io/category/ai-deck/>



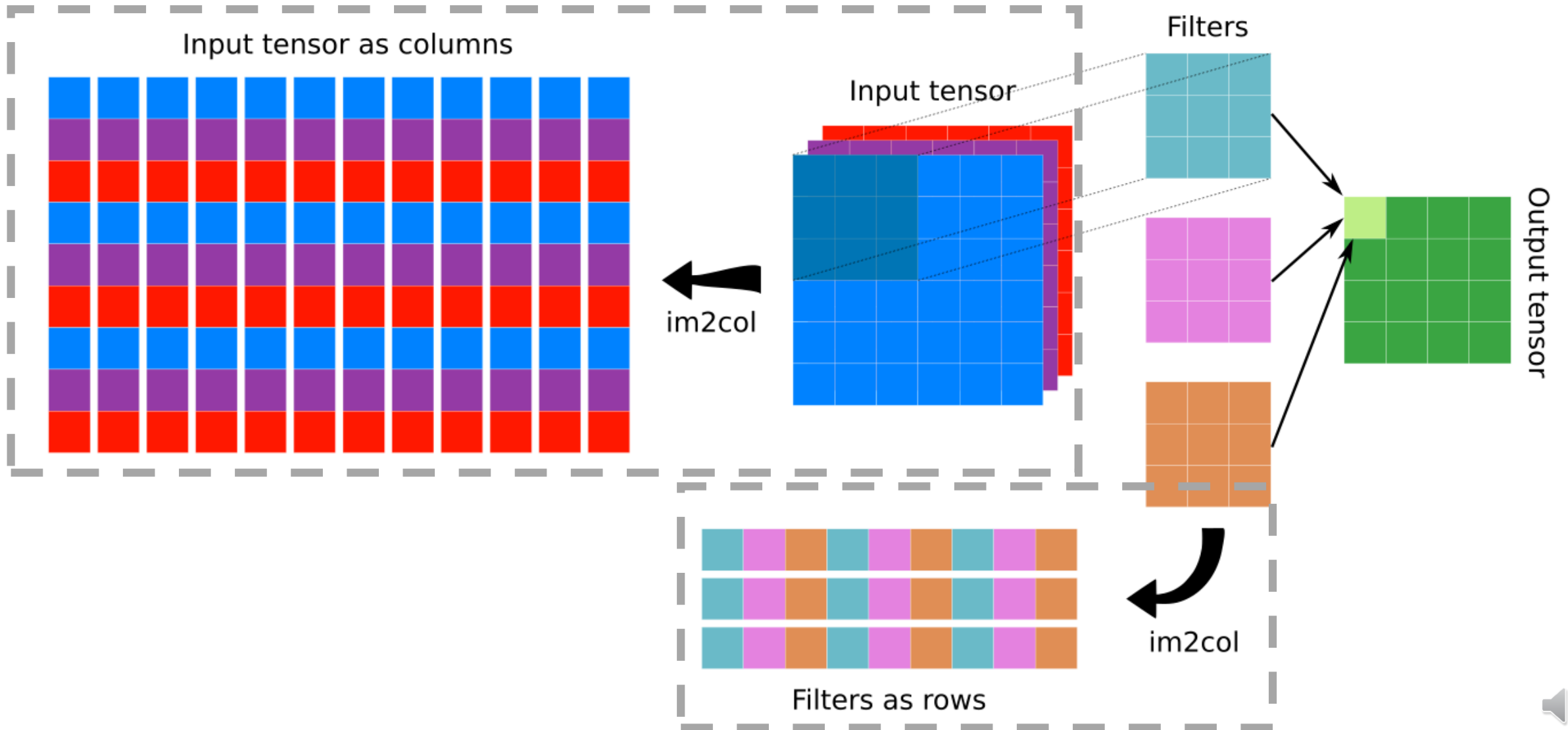
The GAPFlow Tool-set



Convolutions with GAPFlow

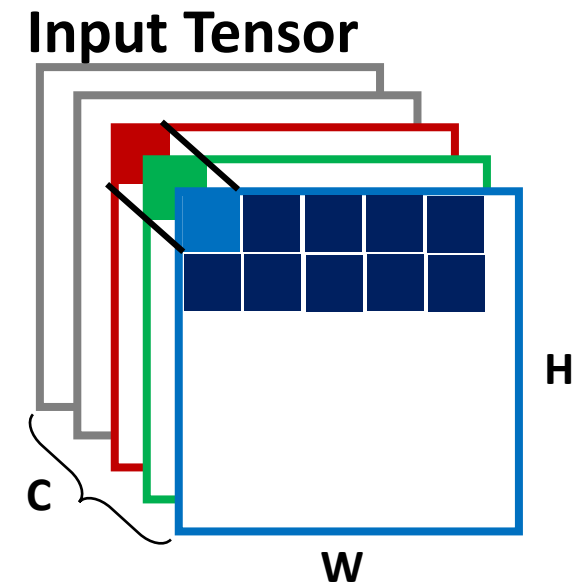


Convolutions with Nemo/Dory

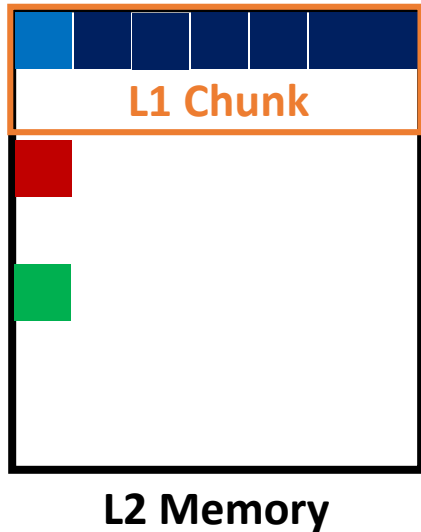


CHW and HWC data layouts

Feature map = the output of one filter applied to the previous layer
Channel = the depth of the input tensor



AutoTiler

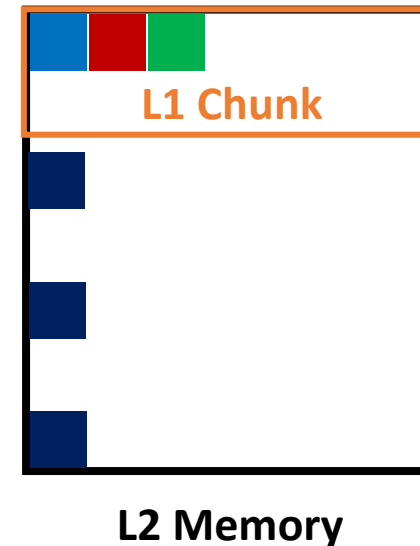


CHW: Channel-last data layout

- 2D activation maps & kernels are stored contiguously
- More efficient on direct convolutions when W or $H > C$.

Size of L1: 64 kB

DORY



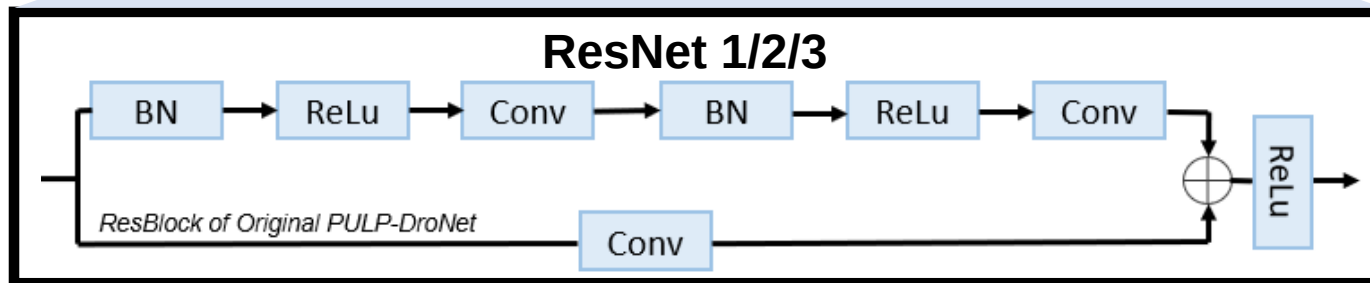
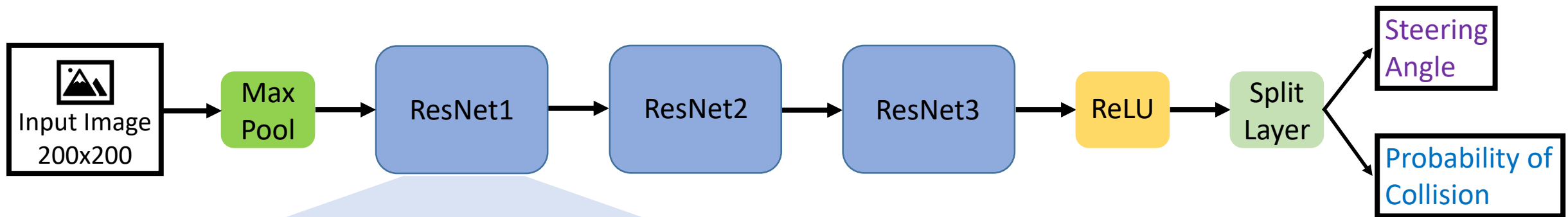
HWC: Channel-first data layout

- The channels of activations are stored contiguously
- Allows to process im2col reordering more efficiently
- More efficient on matrix-matrix multiplication when $C > W$ or H



Architecture of PULP-DroNet

- PULP-DroNet: a vision-based CNN for autonomous driving, based on three ResNet blocks
- The double output is used to command the nano-drone.
- It uses 16-bit quantization.



Two outputs:

- **Steering angle**: regression problem
- **Probability of collision**: classification problem



The Dataset and CNN Topologies

- All three models use the same dataset.
- To simplify quantization:
BN-ReLU-Conv → Conv-BN-ReLU

Dataset

Udacity: steering angle
39.1K images [1]

Bicycle: collision prob.
32.2K images [2]

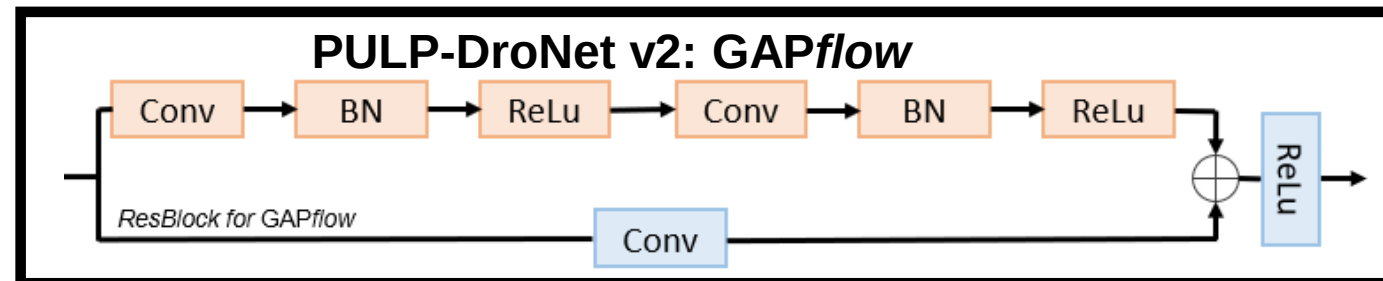
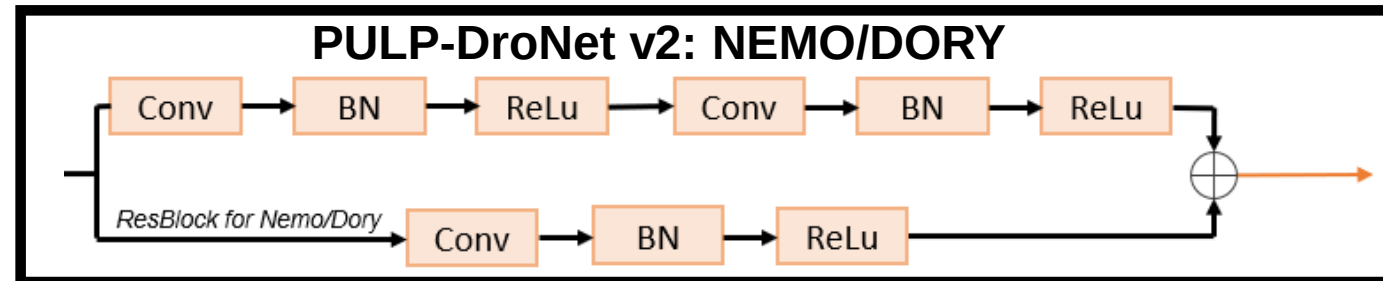
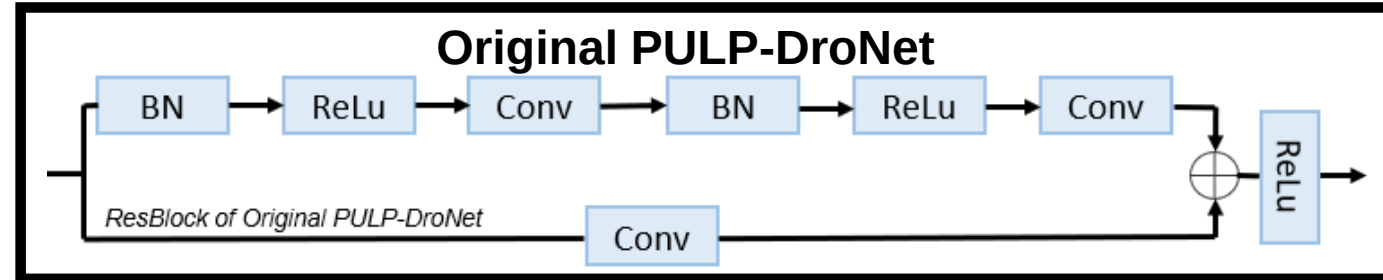
Himax: collision prob.
1.3K images: from the
camera aboard our
nano-drone [3]

Original Dataset:

- Training
- Validation
- Testing

Himax Dataset:

- Training
- Testing



[1] <https://github.com/udacity/self-driving-car/tree/master/datasets>

[2] <http://rpg.ifi.uzh.ch/dronet.html>

[3] <https://github.com/pulp-platform/pulp-dronet>



Experimental Results: Accuracy

- We train the CNN model using the **Original+Himax Dataset**.
- Both pipelines preserve well the performance of the Original PULP-DroNet.
- On the Himax Dataset, the inference performance is fully preserved.

Training		Testing		
		Original Dataset		Himax Dataset
		RMSE	Accuracy	Accuracy
Original Pulp Dronet	Float32	0.109	0.964	0.900
	Fixed 16-bit	0.110	0.977	0.891
Nemo/Dory	Float32	0.118	0.893	0.905
	Fixed 8-bit	0.120	0.892	0.900
GAPflow	Float32	0.136	0.925	0.881
	Fixed 8-bit	0.135	0.925	0.886

Best fixed 8-bit results in bold



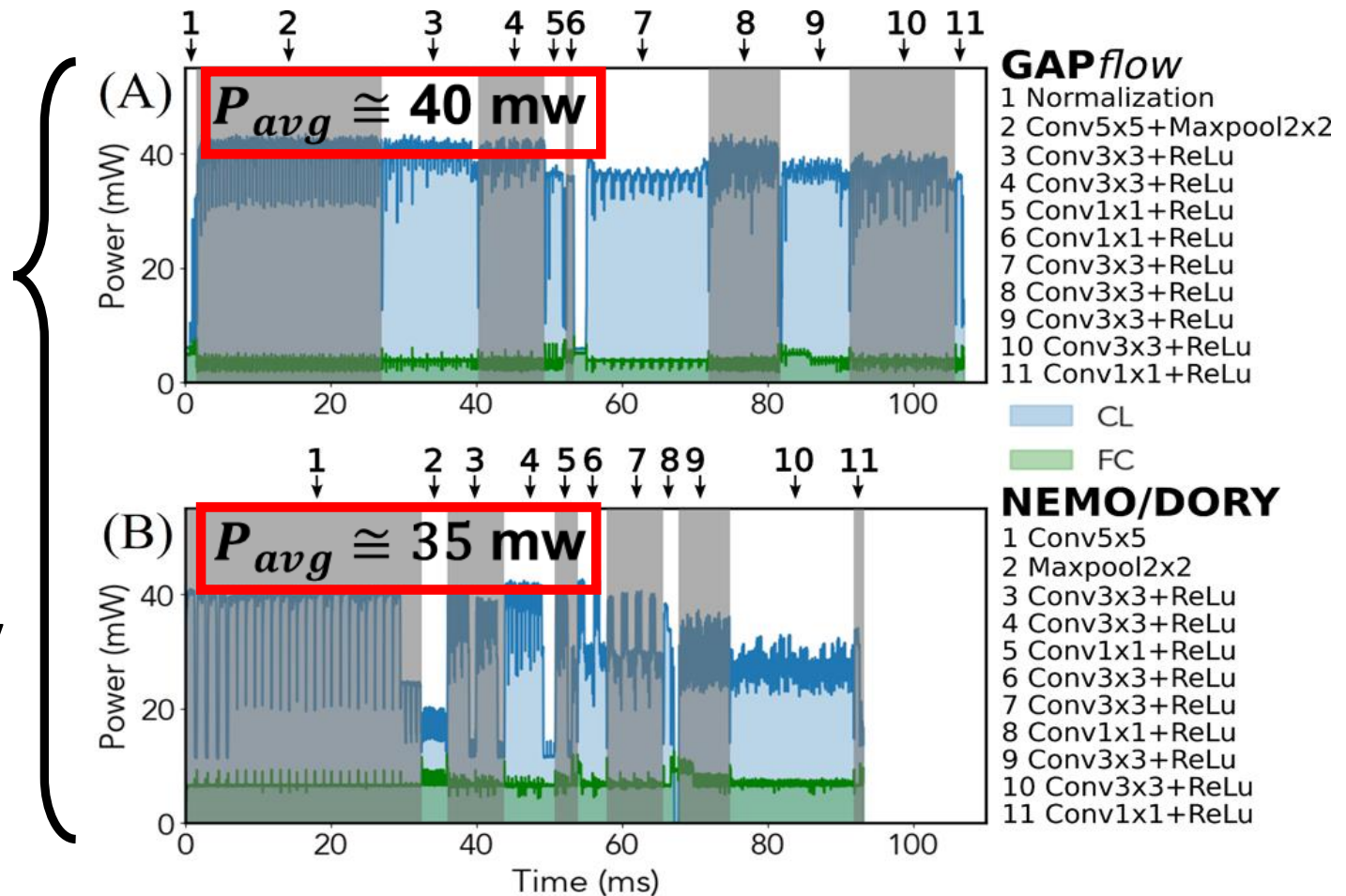
Experimental Results: Energy per framer

Most energy efficient configuration:

- FC@50MHz, CL@100MHz, VDD=1V
- Power waveforms shown in the plot.
- Average energy: 4.4mJ
- Throughput: 10 frame/s

Highest performance configuration:

- FC@250MHz, CL@175MHz, VDD=1.2V
- Average energy: 7mJ
- Throughput: 17 frame/s



GAP8 vs. ARM Cortex M7 with CMSIS-NN Optimizations – analysis based on [1]:

CNN tools	CMSIS-NN [1]		NEMO/DORY
MCU	STM32F746ZG	STM32H723VE	GAP8
# cores	1	1	8
Vdd (V)	1.7	1.4	1.2
Max freq (MHz)	216	550	175 (CL)
MAC/Cycle	0.7	0.7	8.2
MMAC/s	142	390	1435
Power (mW)	221	203	102

Both pipelines are significantly more energy-efficient than ARM-based SoA alternatives

Peak performances

Note: Results for CMSIS-NN are estimated from [1].

STM32 + CMSIS-NN

GAP8 + DORY

Result: GAP8 has

- Faster inference (MMAC/s)
- Lower power consumption



Conclusions

- This work discusses methodologies and tools to automate the complexity reduction and hardware-aware deployment.
- Energy consumption: 3-4mJ/frame.
- Improvement compared to SoA: 2X in memory saving, 1.6X in throughput.

CNN	Quant.	Throughput	MAC	Memory
PULP-DroNet v1	16-bit	6 frame/s	~41M	640kB
v2 (NEMO/DORY)	8-bit	10 frame/s	~41.5M	320kB
v2 (GAPflow)	8-bit	9 frame/s	~41M	320kB



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